Meeting guide (private)

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Goal: show what the residual stream adds (his suggestion), and leave with one decision: generalise first, or go deeper.

30-minute plan

0–3 The question in one line: do late heads "rest" on BOS as a no-op, and what wakes them? Hand him the brief.

3–10 **Result 1** (attention): the matching capital lowers resting at every distance; same-country heads from layer 0; L11H8 is not a previous-token head.

10–20 **Result 2** (residual stream, the main part): the match enters T's stream at layer 1, via L0H2 and MLP0, as a "matching country attended" signal; patching shows the resting effect is assembled in layers 3–7, and from layer 8 the stream alone reproduces it.

20–25 **Result 3**: in the writes, the no-op is a constant offset from BOS attention, not quieter reading. Then the main caveat: 11 pairs.

25–30 **Decide**: generalise (more pairs, cities and demonyms) or go deeper (MLP0, the layer 3–7 assembly)?

Everything at a glance

Work	Main takeaway	Status
Exp 1, baseline	About 74% of heads park on BOS on random tokens, more in late layers. 134 of 144 shift with the token class.	solid
Exp 1ext	The shift comes from the whole context; a single swapped token matters only right before T.	solid
Exp 2, iterations 1–3; Tests A and C	A coherent phrase raises late BOS mass for 12 of 12 phrases; the word pair ending at 31 carries about half. The effect holds for other phrase types, except adjective phrases on rare filler.	solid
Exp 3, relations and controls (8–15)	Completion at the query lowers resting at any distance (R1-in, sweep, E3.1, E3.2). Truth before an unrelated token doesn't. Copy heads take a separate route off BOS. Resting rises with the amount of real text. Several predictions failed.	mixed
Exp 4, controlled difference (random T)	L11H8 reads countries, with recency, and is not a previous-token head. Only L4H11 is a pure previous-token head. The other previous-token heads read nearby words. Four late heads attend at any distance (L10H3 to named entities, L11H7 partly to frequency).	solid
Exp 4b, capital as T	The matching capital lowers late resting at every distance (-0.02 to -0.04). Same-country heads from layer 0 (L0H2, L4H10, L3H0, L1H11). Previous-token heads look away from an adjacent same-country token.	solid
Topology, 4 analyses	Head families are real but not tight clusters. There are no loops, and depth gives only a weak order. Curve shapes are now measured. Only the matching capital closes a layer-0 attention loop (raw attention).	mixed
Exp 5, residual stream (R1–R5, P)	The match enters at layer 1 (R1-b; MLP0 56–59%). It persists to the end, diluted (R4). The resting effect is assembled in layers 3–7 (P-a). No-op = constant offset (R2). Distance dependence rises, not falls (R3 FAIL). Routes depend on weighting (R5 FAIL). Four heads carry 18–36% (P-b, fragile).	mixed

Say this, not that (verified wording)

Say	Avoid
"consistent across rows for these 11 pairs"	"general" / "robust across countries"
"same-country association heads"	"capital lookup heads"
"a matching-country-attended signal"	"the stream encodes which country"
"exact per-pair null, $p = 1/1024$, margin 1.3–1.6×"	"9.7× the null" (that null was too lenient)
"no-op = attention reweighted onto a constant BOS vector"	"resting heads are quiet / read less"
"from layer 8 the stream reproduces the effect within $\pm 6\%$; context still matters at layers 6–7"	"the context doesn't matter after layer 8"
"F rises with layer in every fixed-window variant"	"F rises" with no mention of the shrinking window

Likely questions

Why pair by capital (France … Paris vs China … Paris)? With the same country and different capitals, everything before T is identical, so patching T's stream would trivially carry 100%. Pairing by capital keeps T identical: the stream before layer 0 is equal by construction, which is a built-in null.

Isn't the layer-1 signal just L0H2 moving attention off BOS? Partly. The attention part (about 40%) is a shared BOS→country write that doesn't say which country. But it fires only for the matching country: L0H2 gives 0.34–0.43 to its own country against 0.03 for a wrong one. And 56–59% of the signal comes from MLP0.

How do you know it isn't country identity? The country-identity part of the mean is about 0%. A mismatched country looks like filler to T. A placebo comparing two wrong capitals gives a much weaker signal, below the per-pair null.

Only 11 pairs? Yes, and in paired tests the effective n is 11. We switched to a per-pair null for that reason. Generalising is the main open item, and the first decision to make.

Does topology add anything beyond probes? Honestly, mostly confirmation. The new pieces are the attention-graph loop that only the matching capital closes, and the measured curve shapes (L3H0 prefers far countries). Clustering topology is fragile: Rips and single linkage chain points together.

What does F $\approx$ 1 from layer 8 mean? Patching T's stream never changes earlier positions, so F $\approx$ 1 means late resting depends on T's stream, given context keys that still differ. That isn't trivial. Per layer, the keys still add 2–17%.

Why did the distance test fail? The stream's distance dependence rises with depth, driven partly by the adjacent position. Excluding it, the rise holds on rare filler and is weak on common. With a random final token the predicted fall does appear.

How sure are these numbers? Every analysis was pre-registered with fixed pass rules. Each was re-checked independently: recomputed, Hugging Face re-implementations for patching and the extraction, adversarial verifiers. The verifiers corrected wording in every round.

Next-step options to put to him

A, generalise	More country–capital pairs, plus cities and demonyms at τ. Tests whether the layer-1 signal is "capital of" or same-country relatedness, and replaces 11 with a real sample.	$\sim$1 day
B, MLP0	Split MLP0's 56–59% of the layer-1 signal into neurons. Is it a relatedness detector?	$\sim$1 day
C, layers 3–7	Path patching per head and MLP in layers 3–7, where the resting effect is assembled.	1–2 days

My recommendation: A first. Every result so far is "consistent for 11 pairs", and B and C are more convincing on a real sample.